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CLOSED CORONAL STRUCTURES. II. GENERALIZED HYDROSTATIC MODEL

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ABSTRACT

We use numerical computations of stationary solar coronal loop atmospheres to extend previous analytical work by Rosner, Tucker, and Vaiana. The two classes of loops examined include: (i) symmetric loops with a temperature maximum at the top, but now having length L greater than the pressure scale height s_p, and (ii) loops which have a local temperature minimum at the top. For the first class we find new scaling laws, similar to those found by Rosner, Tucker, and Vaiana, which relate the base pressure p_0 and loop length to the base heating E_0, the heating deposition scale height s_H, and the pressure scale height:

$$T \approx 1.4 \times 10^3 (p_0 L)^{0.33} \exp\,[-0.04L(2/s_H + 1/s_p)] \qquad \text{and}$$

$$E_0 \approx 10^5 {p_0}^{1.17} L^{-0.83} \exp\,[0.5L(1/s_H - 1/s_p)]\,.$$

Loops for which L is greater than $\sim$ 2 to 3 times the pressure scale height do not have stable solutions unless they have a temperature minimum at the top. Computed models with a temperature inversion at the top are allowed in a wider range of s_H values than are loops with $T_{\max}$ at the top. We discuss these results in relation to observations showing a dependence of prominence formation and stability on the state of evolution of magnetic structures, and we suggest a general scenario for the understanding of loop evolution from emergence in active regions through the large-scale structure phase to opening in coronal holes.

Subject headings: hydromagnetics — Sun: corona

I. INTRODUCTION

The fact that the solar outer atmosphere is quite structured has been long recognized (cf. van de Hulst 1953), but it was not until the advent of X-ray imaging of the Sun, and especially as a result of the ATM mission, that the role of the magnetic field connectivity ("open" or "closed") in spatially modulating the coronal plasma temperature and density was firmly established (Krieger, Timothy, and Roeloff 1973; Noci 1973; Pneuman 1973; Vaiana, Krieger, and Timothy 1973). The hot plasma is essentially confined by the magnetic field into bright arches or loops which have their footpoints in the chromosphere and which undergo only slow evolution (apart from some transient phenomena, such as flares) over a time scale of several solar rotations.

The basic recognition of the loop structuring and of the role of the magnetic fields has motivated a strong effort to revise existing models of the corona, traditionally based on the picture of an homogeneous stratified atmosphere. Rosner, Tucker, and Vaiana (1978, hereafter RTV) have derived, by means of a simple model for the confined plasma, scaling laws between length of the loops and temperature and pressure of the confined plasma. These scaling laws, although derived under restrictive conditions (e.g., constancy of pressure and heat deposition along the loop), reasonably well describe the observed correlation of these parameters. Hood and Priest (1979) have generalized these scaling laws to include the possibility of substantial conductive losses to the chromosphere. Vesecky, Antiochos, and Underwood (1979) have studied with a numerical model the effect of the loop geometry on observable parameters such as the X-ray brightness of the loops. On the other hand, the correlation between magnetic field strength and X-ray emission (Golub *et al.* 1980) strongly suggests that magnetic field–related heating mechanisms should be considered (Vaiana and Rosner 1978; Wentzel 1978), and some suggestions along these lines have indeed been made (Tucker 1973; Rosner *et al.* 1978; Ionson 1978; Galeev *et al.* 1980; Habbal, Leer, and Holzer 1979).

No attempt, however, has been made to include in this picture the evolution of magnetic fields from the emergence in active regions to their slow diffusion. Existing models do not explain, for instance, the preferential formation of prominences during an intermediate phase of loop evolution (Serio *et al.* 1978) or the relation between large-scale structures (tenuous loops with sizes 10^{10} cm and over) and coronal holes (large zones devoid of X-ray emission).

In order to clarify these problems, we have taken the approach of building numerical models of static coronal loops along the lines of RTV but removing some of their constraints in order to allow pressure and heating to vary along loops. Although these models are static, the study of the dependence of the solutions on those parameters which are related to the evolutionary paths followed by coronal loop structures may give insight for the understanding of loop stability and evolution (see also § VI for discussion).

Static modeling is employed not because we believe static conditions are prevalent in the corona; rather, the types of X-ray emitting structures we model (ranging from active region to quiet Sun loops) are observed to be relatively quiescent on typical cooling time scales so that the static approximation ought to be reasonable for the coronal atmosphere.

In the next section (§ II) we shall introduce the model and the parametrization used. In § III we shall derive new more general scaling laws for the heating function and the temperature of the loops. A presentation and a discussion of the numerical results for simple loops having temperature maxima at the top will be given in § IV, while results for loops with temperature minima on top will be given in § V. An overview of how our results relate to a general framework of coronal loop evolution will be given in § VI.

II. STATIONARY LOOP MODEL

The basic physics of a confined atmosphere can be understood as follows. The stationary equations of motions for an extended atmosphere define a boundary value problem, which in the case of a plane-parallel (unconfined) atmosphere reduces to obtaining a solution subject to specified conditions at the solar surface ($r = R_0$) and at infinity. The basic property of this solution is that the position of the coronal temperature maximum r_m [$\equiv r(T_{\max})$] is functionally related to the value of the maximum coronal temperature $T_{\max}$, e.g.,

$$r_m = f(T_{\max}) \tag{2.1}$$

(see, for example, Hearn 1975); thus, for example, a hotter atmosphere (for given base pressure) implies a more extended atmosphere.

This degree of freedom is absent in the case of loop (confined) atmospheres. That is, for a loop atmosphere $r_m \leq L$, where L is the distance between the loop footpoint and apex (assuming a symmetric loop); this constraint results in a coupling of loop temperature and pressure for temperatures sufficiently high that $r_m = L$. The above line of analysis led RTV to derive parameter-free scaling laws for the loop coronal temperature and local heating rate under the restrictive assumption that the pressure $p \sim$ constant (e.g., restricted to loops whose size is small or comparable to their pressure scale height):

$$T_{\max} \approx 1.4 \times 10^3 (pL)^{0.3} K , \tag{2.2}$$

$$E_H \approx 10^5 p^{1.17} L^{-0.8} \text{ ergs cm}^{-3} \text{ s}^{-1} , \tag{2.3}$$

where E_H is the rate of heat deposition in the loop, assumed to be uniform. In this model the magnetic field has the sole function of defining the shape of the loop and confining the plasma.

The basic equation of the RTV model is the energy equation

$$E_H + E_R = \text{div } F_c , \tag{2.4}$$

where E_R is the energy radiated from the loop, and F_c is the thermal conductive flux. By defining a coordinate s along the loop's length, we have

$$F_c = -\kappa T(s)^{5/2} \frac{dT(s)}{ds} , \tag{2.5}$$

where κ is 9.2×10^{-7} ergs s^{-1} cm^{-1} K^{-1}.

We want to study this loop model by relaxing the condition of constant pressure; therefore we shall follow the approach of Rosner and Vaiana (1977) by considering the equation for hydrostatic equilibrium, which, in the assumption of static (no flow) and free (no external forces other than the gravity) loops, assumes the form (see Rosner and Vaiana 1977 for the basic assumption of this model)

$$\frac{dp(s)}{ds} = \frac{\mu m_{\rm H}}{2k} \frac{p(s)}{T(s)} g_s , \tag{2.6}$$

where μm_H is the average ion mass, k the Boltzmann constant, g_s the component of gravity along the loop, and $p(s)$ the pressure, which is related to the plasma density $n(s)$ by

$$p(s) = 2n(s)kT(s) . \tag{2.7}$$

The radiative losses per unit volume are given by $-p^2 P(T)/4k^2T^2$ (Tucker and Koren 1971), and we will use the parametrization of $P(T)$ given by RTV. We have parametrized the heating function as follows:

$$E_H = E_0 \exp\left[-s/s_H\right] , \tag{2.8}$$

where E_0 is the heat deposition at the base of the loop, and s_H is a free parameter which determines the heating scale height. Rosner, Tucker, and Vaiana (1978) have pointed out that loop configurations for which the temperature maximum is below the apex of the loop might result in the formation of stable prominences, provided that the cooler material in the upper part of the loop has some means of mechanical support. In order to address this issue without the complication of a detailed model of prominence supporting magnetic fields, we have investigated models in which the region above the temperature maximum is constrained on a horizontal or vertical tube. We believe such a configuration to be a reasonable first approximation to the true geometry (cf. Serio *et al.* 1978); moreover, this geometry is amenable to numerical simulation and should give insight into the characteristics of the solution in a more realistic case.

Vesecky, Antiochos, and Underwood (1979) have shown that in a loop model of the kind we have studied, the conditions in the higher portion of the loop are insensitive to the conductive flux one uses as a boundary condition at the loop footpoint. The same result was discussed in Galeev *et al.* (1980). We shall not, therefore,

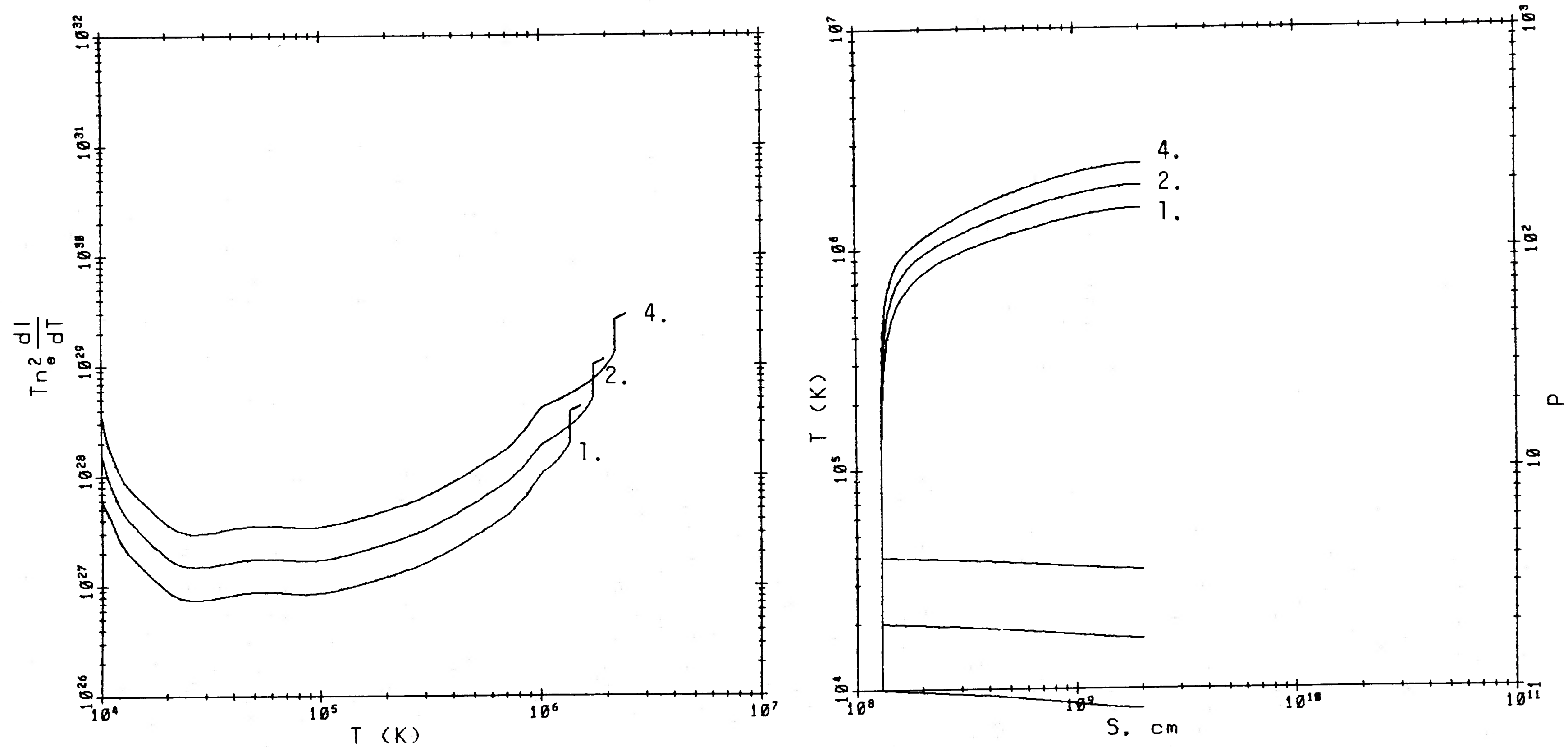

FIG. 1.—Dependence of the differential emission measure (in cm^{-5}), temperature (in K), and pressure (in dynes cm^{-2}) on various model parameters, for a small loop having $L = 2 \times 10^9$ cm. (*a*) Effect of varying the base pressure from 1 to 4 dynes cm^{-2}; loop parameters used are listed in lines 1, 2, and 3 of Table 1. Top box shows the differential emission measure as a function of temperature and bottom box shows temperature T and pressure p as functions of distance s along the loop.

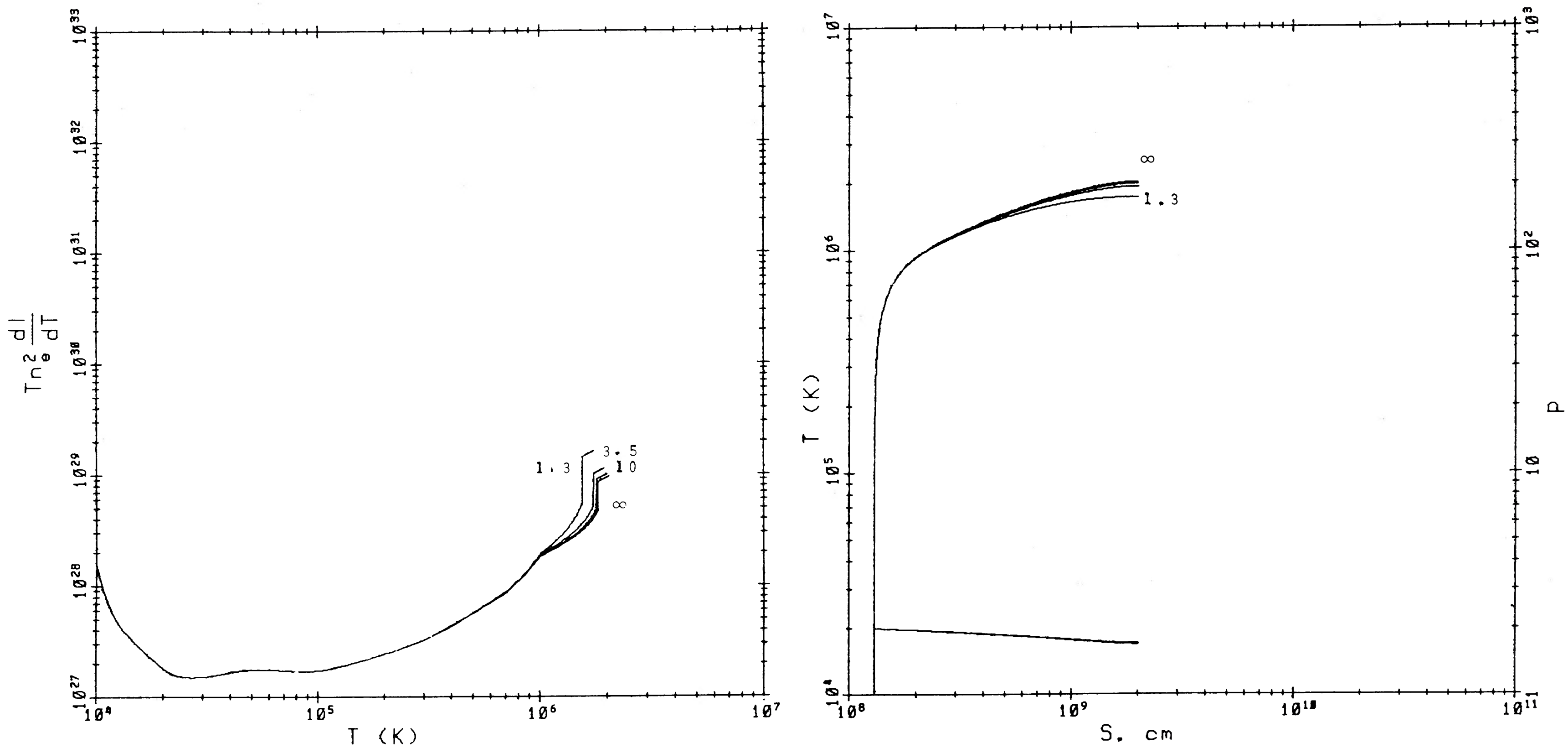

FIG. 1.—(*b*) Effect of varying the heating scale height, from 1.3×10^9 cm to ∞ on differential emission measure (*top box*) and T and p (*bottom box*); loop parameters are listed in lines 2, 4, 5, and 6 of Table 1.

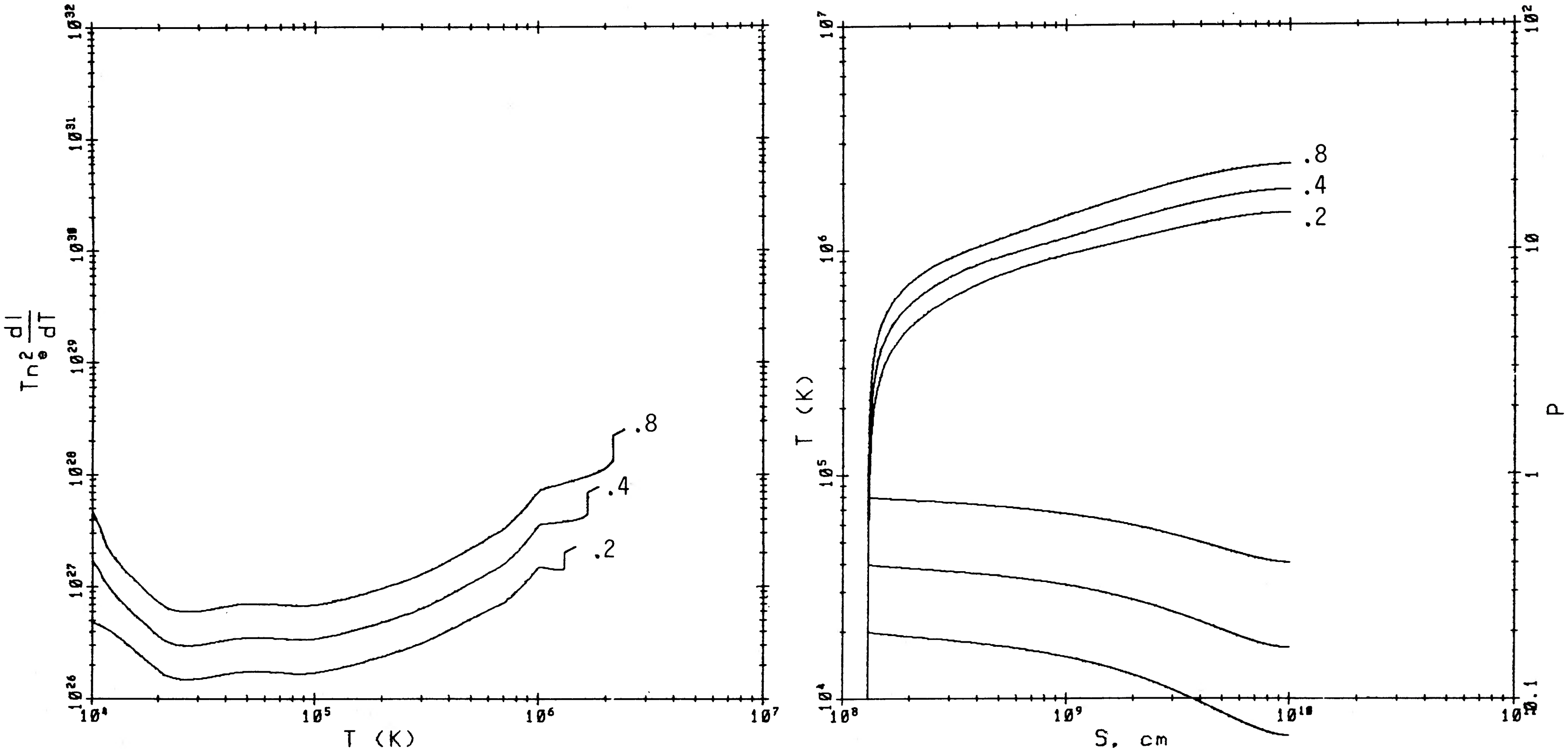

FIG. 2.—As in Fig. 1, but for a loop having $L = 1 \times 10^{10}$ cm. (*a*) Effect of varying the base pressure, from 0.2 to 0.8 dynes cm^{-2}; lines 7, 8, and 9 of Table 1.

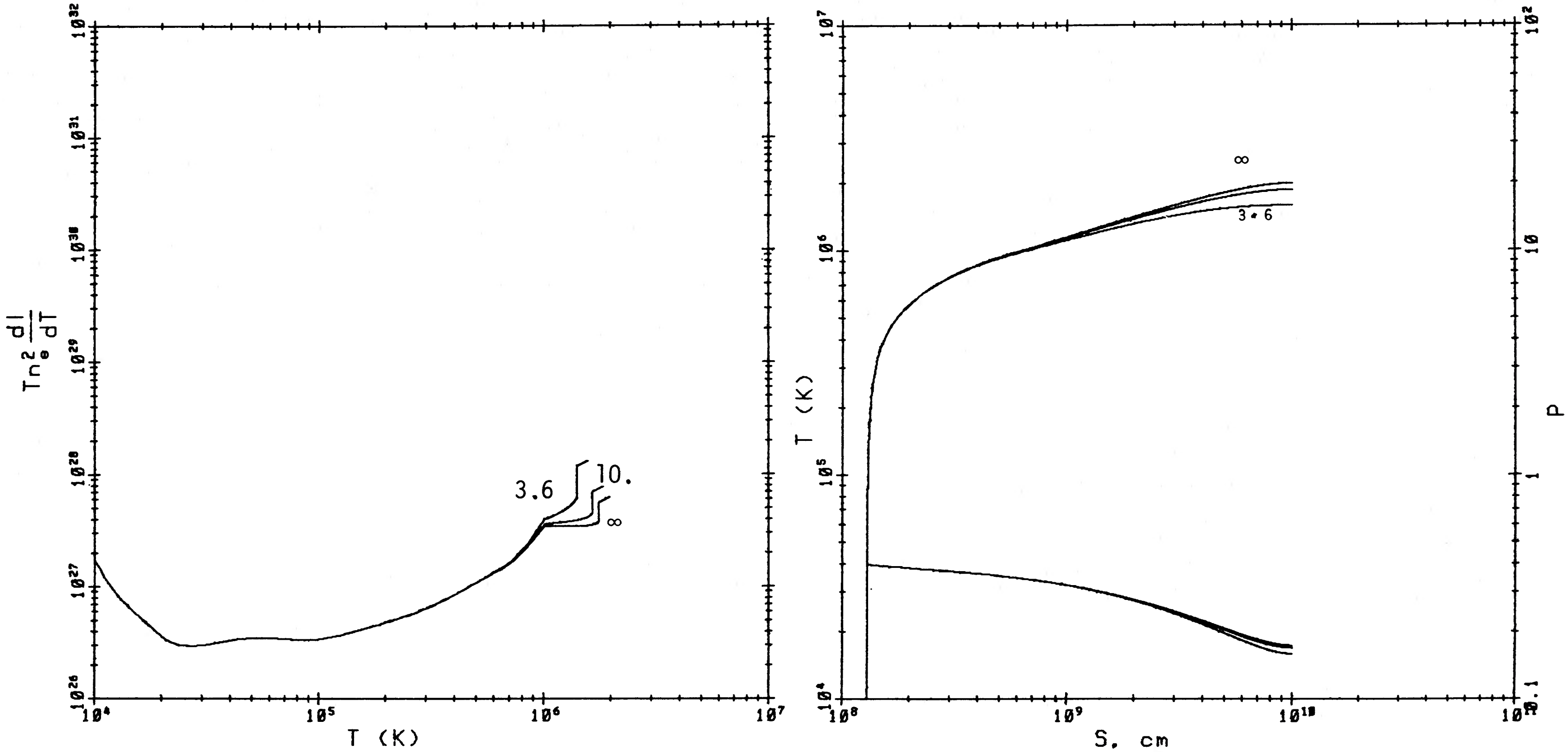

FIG. 2.—(*b*) Effect of varying the heating scale height from 3.6×10^9 cm to ∞; lines 8, 10, and 11 of Table 1.

address this issue here but shall give a brief discussion of the effect of varying the loop cross section, also discussed by Vesecky, Antiochos, and Underwood (1979).

Our algorithm for the solution of equations (2.4) and (2.6) uses the Runge Kutta method to iteratively determine the base heating E_0 given the loop length L and its base pressure and using a variable integration step determined by imposing $\Delta T/T < 0.01$ at each step.

III. SCALING LAWS FOR LARGE CORONAL LOOPS

The requirement for energy balance in the loop and the condition that the temperature maximum be at the top of the loop determine relationships among the parameters of the loop model, namely the temperature, length, pressure, and heating function. RTV and others have derived scaling laws for loops with uniform pressure. To see how these scaling laws can be analytically extended to large loops, i.e., loops whose size is comparable to or larger than the typical coronal pressure scale height, we proceed along the lines of RTV, but also take into account the possible spatial variation of pressure and heat deposition (eqs. [2.6] and [2.8]). We first note that because the transition region between chromosphere and corona is very thin, $p \sim$ constant across it; the dominant pressure drop thus occurs at coronal temperatures, so that, to a good approximation, equation (2.6) can be integrated by assuming $T \sim$ constant, e.g.,

$$p(s) \sim p_0 \exp\left[-s/s_p\right], \tag{3.1}$$

where p_0 is the loop base pressure and $s_p[= 2kTm_\mathrm{H} g_0 \approx 5 \times 10^3 T \text{ cm}]$ is the pressure scale height.

The energy equation can be integrated in the form

$$F_c^{\,2}[T(s)] = f_R[T(s)] - f_H[T(s)], \tag{3.2}$$

where

$$f_R[T(s)] = \int_{T_0}^{T(s)} \frac{\kappa p(s)^2 T'^{1/2}}{2k^2} P(T')\, dT', \tag{3.3}$$

$$f_H[T(s)] = \int_{T_0}^{T(s)} 2\kappa E_0 T'^{5/2} \exp\left[-s(T')/s_H\right] dT'. \tag{3.4}$$

We can estimate the values taken on by f_R and f_H at the top of the loop where we set $T(L) = T$, taking straightforward application of the mean value theorem

$$f_R(T) \sim \frac{\kappa P_0^{\,2}}{2k^2} \exp\left[-\beta L/s_p\right] \int T^{1/2} P(T)\, dT, \tag{3.5}$$

$$f_H(T) \sim \tfrac{4}{7}\kappa E_0 \exp\left[-\alpha L/s_H\right] T^{7/2}, \tag{3.6}$$

where α and β are constants to be determined.

If we now proceed as in RTV, we can derive scaling laws of the form

$$T \approx 1.4\; 10^3 (p_0 L)^{0.33} \exp\left[-\alpha' L/s_H - \beta' L/s_p\right], \tag{3.7}$$

$$E_0 \approx 10^5 p_0^{\,1.17} L^{-0.83} \times \exp\left[(\alpha + \tfrac{5}{2}\alpha')L/s_H - (\beta - \tfrac{5}{2}\beta')L/s_p\right]. \tag{3.8}$$

We have determined the parameters α, β, α', β' by comparing equations (3.7) and (3.8) to the results of numerical calculations for loops ranging in height from $\sim 10^8$ cm to 4×10^{10} cm and for values of s_H in the range to be discussed below. We find

$$\alpha' = -0.08,$$

$$\beta' = -0.04,$$

$$\alpha + \tfrac{5}{2}\alpha' = \beta - \tfrac{5}{2}\beta' = 0.5.$$

These scaling laws can be used to set a limit to the possible allowed values of s_H by noting that the left-hand side of equation (2.4) must be positive at $s = L$ if the loop is to have a temperature maximum at the top. We thus have

$$E_0 e^{-L/s_H} \gtrsim \frac{p_0^{\,2} \exp(-2L/s_p)}{4k^2 T^2} P(T), \tag{3.9}$$

which, by use of (3.7) and (3.8), and of the fact that α', $\beta' \ll 1$, reduces to

$$\exp\left[-\frac{L}{2s_H} + \frac{1.5L}{s_p}\right] > 1.8 \times 10^{18} P(T)(T)^{1/2}. \tag{3.10}$$

At coronal temperatures we find, for example, that for $L \ll s_p$, $L \lesssim 2s_H$. The exponential decrease with height of the pressure, as implied in (3.9), has been numerically tested on a variety of loops, the actual dependence always being within 20% of $p_0 \exp[-s/s_p]$, where s_p is the pressure scale height corresponding to the maximum temperature in the loop.

IV. RESULTS FOR LOOPS WITH MAXIMUM TEMPERATURE AT THE TOP

The numerical approach which has been described in § II is designed for detailed modeling of observed loop atmospheres, to calculate expected EUV line and X-ray broad-band emission together with the radio brightness temperatures at various frequencies. The basic input parameters in the model are the loop length L, the base heating rate, heating scale height s_H, and the base pressure p_0. An additional input parameter of interest describes the field geometry near the loop footpoints, e.g., whether the flux tube is constricted in the transition region, and how large the constriction factor is (cf. discussion of this point in Rosner and Vaiana 1977). The results of our calculations for two loops of different heights (5×10^9 cm and 2×10^{10} cm) are shown in Figures 1 and 2, and tabulated in Table 1; a discussion of these calculations follows below.

a) Variation in Base Pressure

Our numerical model substantially verifies and extends the parameter free scaling law between pressure, temperature, and loop length, originally derived by RTV in the case of uniform heating and constant pressure, to loops whose size is larger than the pressure scale height and the heating scale height. The result we obtain was anticipated by data for large-scale structures shown in RTV, namely that the scaling law remains unchanged, but one must use the *base* pressure in evaluating the scaling laws. Our equation (3.7) is substantially the RTV scaling law, since α' and β' are much less than 1.

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TABLE 1

PARAMETERS USED IN THE LOOP MODELS SHOWN IN FIGURES 3 AND 4

Model (1)	Base Pressure (dyn cm^{-2}) (2)	Length (cm) (3)	Tube Expansion (4)	Heating Scale (cm) (5)	Base Heating (ergs cm^{-3}) (6)	Maximum Temperature (K) (7)
1	1.0	2×10^9	5	3.5×10^9	1.69×10^{-3}	1.52×10^6
2	2.0	2×10^9	5	3.5×10^9	4.27×10^{-3}	2.00×10^6
3	4.0	2×10^9	5	3.5×10^9	1.02×10^{-2}	2.44×10^6
4	2.0	2×10^9	5	∞	3.11×10^{-3}	2.05×10^6
5	2.0	2×10^9	5	1×10^{10}	3.50×10^{-3}	2.00×10^6
6	2.0	2×10^9	5	1.3×10^9	7.19×10^{-3}	1.74×10^6
7	0.8	10^{10}	5	1×10^{10}	3.39×10^{-4}	2.41×10^6
8	0.4	10^{10}	5	1×10^{10}	1.26×10^{-4}	1.85×10^6
9	0.2	10^{10}	5	1×10^{10}	4.78×10^{-5}	1.46×10^6
10	0.4	10^{10}	5	∞	7.79×10^{-5}	1.98×10^6
11	0.4	10^{10}	5	3.6×10^9	2.55×10^{-4}	1.58×10^6

NOTE.—Date in columns (2)–(5) are input parameters, while those in columns (6) and (7) are derived from the model.

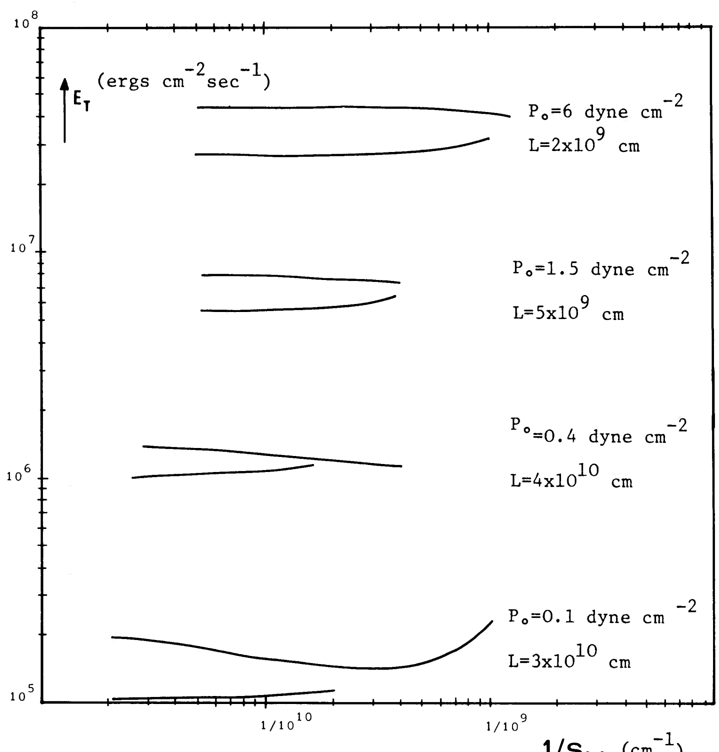

FIG. 3.—Dependence of the total heating (E_T) required to obtain meaningful solutions as a function of the inverse heating scale height. The loops have lengths and pressures as indicated and are typical of active region loops, large-scale structures, and evolved large-scale structures. Class I solutions, which have generally the lowest E_T, have the temperature maximum on top; class II have a local minimum in the temperature on top. Note that for some values of base pressure p_0 and loop length L, there is generally a value of the heating scale height at which the class of solutions having a temperature inversion at the top become lower energy solutions.

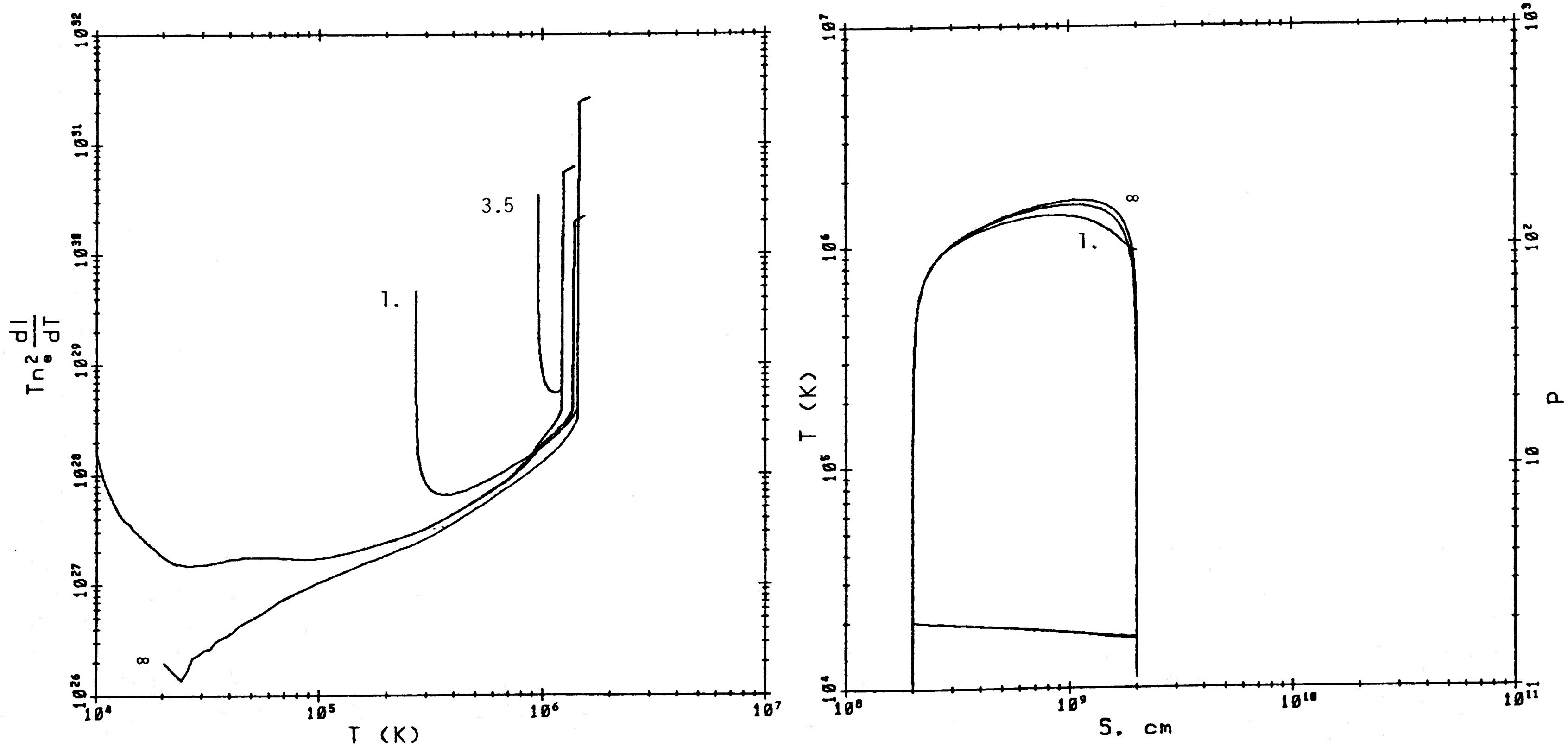

FIG. 4.—Temperature (in K), pressure (in dynes cm^{-2}), and differential emission measure (in cm^{-5}) profiles for loops with a temperature inversion, for various values of the heating scale. Each differential emission profile is split into two branches, corresponding to the section of the loop from the base to the temperature maximum and from the temperature maximum to the local minimum at the top of the loop. The models are terminated below $T = 10^4$ K. (*a*) Loop with base pressure = 2 dyn cm^{-2} and length = 2×10^9, and $s_H = 1 \times 10^9$ cm, 3.5×10^9 cm, and ∞.

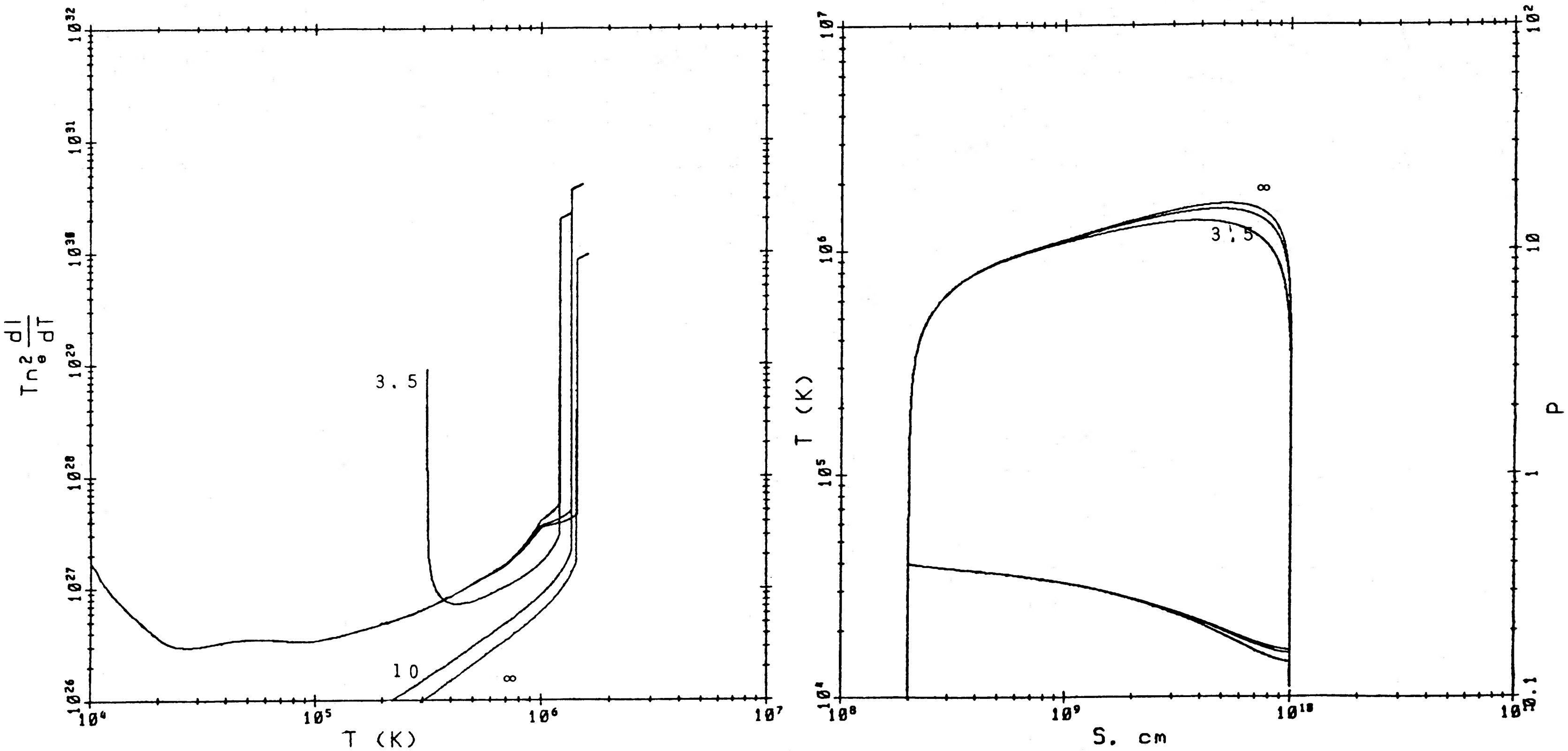

FIG. 4.—(*b*) Loop with base pressure 0.4 dyn cm^{-2} and length 10^{10} cm, for $s_H = 3.5 \times 10^9$ cm, 1×10^{10} cm, ∞.

b) Variation in Base Heating and Heating Scale Height

In the case of uniform heating, once the loop length has been fixed together with the base pressure, the base heating E_0 is completely determined by the RTV scaling law (2.3). In our approach it will instead depend solely on the heating scale height s_H. Since most of the energy requirement of the loop is determined by radiation losses near the base, the total energy, $E_T = \int E_H\,ds$, will approximately be constant, independent of s_H, at least for loops where $L/s_H < 1$. Decreasing s_H has the general effect of decreasing the loop temperature (Figs. 1 and 2) while increasing the differential emission measure. As a result, the total power requirement will increase when $L/s_H > 1$, as can also be inferred from equation (3.8). In this range of s_H, the total energy is actually increasing exponentially as a function of L/s_H.

From the model calculations we also see that we cannot build meaningful solutions for relatively small values of s_H. For loops much smaller than the pressure scale height, we find in fact numerical solutions only for $s_H \gtrsim L/2$, as derived in § III. A similar result was conjectured by RTV on the basis of the thermal stability of the system. The novelty of our result is that in this case solutions do not exist at all, independent of their stability. For loops whose length is greater than the pressure scale height, the limiting value of s_H is roughly given by $s_p/3$. Smaller values of s_H will not give solutions with a temperature maximum at the top. We want to recall also that our model is strictly static and therefore the solutions which we find might not be stationary. We shall discuss this point further in § VI.

c) Variation in Flux Tube Expansion

We have used the form of flux tube expansion in the transition region between the temperatures 7×10^5 K and 10^6 K which was discussed by Rosner and Vaiana (1977). Our results are substantially equivalent to those of Vesecky, Antiochos, and Underwood (1979); the loop temperature and pressure profiles are almost unchanged, and the only appreciable effect is a slight increase in differential emission measure in the above-mentioned temperature range. Notice that in Figures 1 and 2 we have not multiplied the differential emission measure by the area expansion factor as in Vesecky, Antiochos, and Underwood (1979). Their procedure, which is applicable to loops seen on the limb, might be used incorrectly in comparing with observations the computed X-ray surface brightness of high loops seen on the disk, when the line of sight is along most of the loop's length. In this case the base area expansion is in fact irrelevant to the computed X-ray surface brightness.

V. LOOPS WITH A TEMPERATURE INVERSION

A different class of solutions to the energy equation is the one for which the temperature maximum is at some location along the loop, while a temperature minimum is at the top. We shall refer to these loops as class II, while loops with the temperature maximum at the top will be called class I. As discussed in RTV, these solutions will be unstable against Rayleigh-Taylor instability whenever $(\nabla n \cdot g) < 0$ and Lorentz forces provide no mechanical support. We have in fact numerically verified that, despite the exponential decrease of pressure in large loops, there is a substantial portion of the loop with increasing density if a temperature inversion occurs.[1] We have therefore computed loop models whose geometry can support the denser material, as discussed in § II; here we assumed that the magnetic field (which defines the geometry) provides the necessary support. Geometries of this kind, notably of a semicircular loop with a vertical downward bend near the loop apex, have been suggested in several prominence models (viz., Pikel'ner 1971). The principal results (e.g., scaling laws) of numerical calculations for large loops of this kind were found to be quite similar to those of semicircular geometry; therefore, in the following we shall restrict our attention to the latter geometry.

We first notice that for loops having the same length and boundary conditions, the total energy requirement E_T for class II solutions is higher than for the corresponding class I solutions, since more energy is required by the denser and cooler plasma above the temperature maximum. Figure 3 shows how E_T depends on the heating scale height for the two classes of loops, for two different values of base pressure and loop length. Notice that, contrary to class I loops, E_T actually decreases with s_H, and it will in some cases become equal to or less than $E_T{}^{\mathrm{I}}$. Thus a range of values exists for which the energy requirements of a prominence-type loop become less than those of a loop with $T_{\max}$ at the top.

We can locate the position where this crossover occurs by deriving an approximate scaling law for the energy deposition. The approach followed in deriving (3.8) fails here sine the plasma in the upper part of the loop may be very tenuous and still hot, making the approximations which led to (3.7) and (3.8) unrealistic. However, we obtain a reasonable empirical fit to the results of the numerical model with the expression:

$$E_0{}^{\mathrm{II}} \approx 1.8 \times 10^5 p_0{}^{1.17} L^{-0.83} \exp\left[\frac{0.3L}{s_H} - \frac{0.5L}{s_p}\right] \quad (5.1)$$

with $s_H \gtrsim L/4$ or $s_p/4$, whichever is smaller. The point at which $E_T{}^{\mathrm{I}}$ reaches $E_T{}^{\mathrm{II}}$ is therefore given by

$$L/s_H \sim 3\ .$$

To give an idea of the accuracy of (5.2) and (5.1), we point out that from the numerical results or from extrapolations of the model calculations shown in Figure 3 we get $2 < L/s_H < 3$.

We next examine the general character of the class II solutions in the same detail as was done for class I solutions (Fig. 4). We see that, as might have been expected, the largest loops (Fig. 4b) have little emission measure on top despite the increased density. We also see that generally, more uniformly heated loops (e.g., those labeled s_H = " ∞ ") tend to have more emission near the

[1] Except in cases for which the minimum and maximum temperatures differ by only a small amount.

top. Note that the curves in Figure 4 differ from those presented earlier, in that the differential emission measure is no longer single-valued as a function of T. For each value of heating scale height, the calculated emission measure curve reaches a maximum temperature value then returns down to lower temperature as we continue toward the top of the loop.

VI. OVERVIEW AND DISCUSSION

The results of these loop models, although stationary, give insight into the slow dynamics of the loop atmosphere. Our working hypothesis is that much of the observed dynamics—including the presence of topologically open regions (coronal holes)—is the result of the evolution of the loops as the magnetic fields which define their form decay under the influence of randomizing photospheric fluid motion.

The overall evolutionary framework suggested is as follows. The emergence of a photospheric magnetic flux tube above the solar surface is accompanied by rapid plasma heating, which, because of the confining constraint imposed by the magnetic field, enforces a correlation between the loop pressure and temperature. The resulting evolutionary trajectories (Fig. 5) are therefore signatures both of the confined nature of the loop plasma and of the heating process. The form of these trajectories, which indicates a well-defined "corridor of evolution" for the radiative loss (and heating) rates with increasing loop size (and hence with more diffuse magnetic flux) suggests that the coronal magnetic fields—in addition to playing a passive role by spatially confining the loop plasma—is actively involved in defining the loop energy balance.

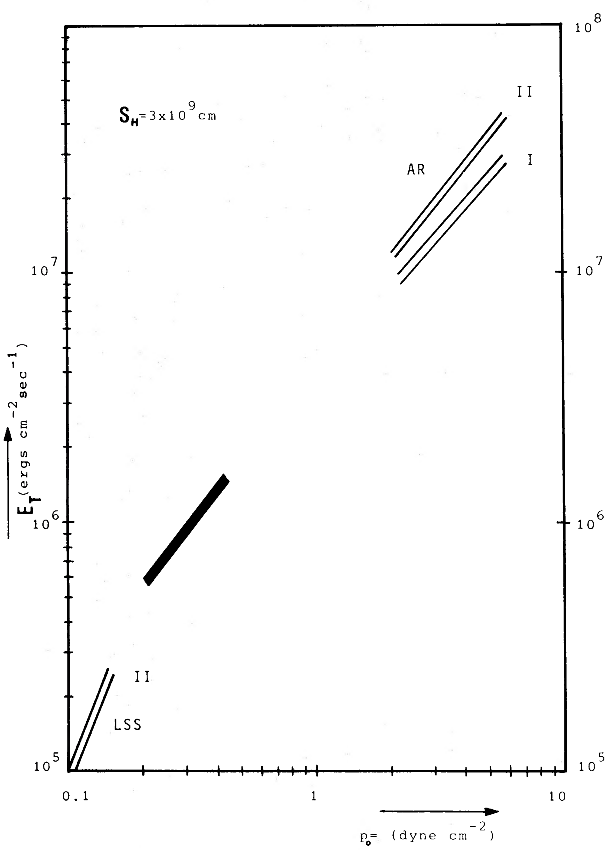

FIG. 5.—Conceptual evolution of characteristic loop plasma parameters, base pressure p_0, and total energy required E_T for heating scale height 3×10^9 cm. In the active region phase, both solutions of class I (T_{max} at top) and II (T inversion along loop) are possible. Class II solutions require more energy during this phase, and they are therefore less stable. In the intermediate phase of evolution, class I and II solutions are close in energy as indicated by the dark band, and from there on only class II solutions exist, until loops eventually open up to the solar wind.

During the *active region* phase, the loop size is less than or comparable to the pressure scale height of the hot plasma it contains; the pressure is therefore fairly constant along its length. Because of this constraint, the active region coronal loop atmosphere may be thought of as "stiff" in the sense that it is strongly determined by the boundary conditions at the chromospheric interface; thus, possible thermal instabilities during this phase are strongly influenced by the loop footpoint boundary layers and are constrained to be isobaric (Habbal and Rosner 1979).

The gradual diffusion of the photospheric fields in active regions leads to a monotonic increase in the size of the overlying coronal loop structures. During the *large-scale structure* (or *quiet Sun*) phase, the loop structures are large when compared with the coronal pressure scale height. In consequence, considerable pressure variations along a given loop can be expected. Because the pressure falloff with height is now exponential while the temperature is fairly uniform, the radiative emission per unit volume ($\propto p^2$) experiences a similar but more drastic decrease with height. This fact has the immediate observational consequence that—in the absence of thermal instabilities—for a similar fractional variation in base pressure, large-scale structure loops will evince far less noticeable fluctuations in coronal emission than active region loops (e.g., coronal emission fluctuation in the quiet Sun will be smaller by a factor $\sim e^{-2L/s_p}$). Thus, even if one assumes the level of pressure fluctuations to be identical at the footpoints of active and quiet Sun loops, the coronal atmosphere in quiet Sun loops will indeed appear to be "quiet." However, the increased loop size implies that loops longer than $\sim 2s_H$ will develop a temperature inversion. We suggest that cool condensations within large-scale loop structures may therefore be present in those loops; for loops much higher than the pressure scale height, however, there is virtually no material on top to be seen, despite the fact that it is cool. This has to be related to observations suggesting that stable prominences are associated with partially decayed active region loops, and that they tend to disappear at a later stage of decay (Serio *et al.* 1978).

We might get an insight into the typical deposition scale for the heating, by considering that hot loops are observed above stable quiescent prominences. By considering that the coronal loops above stable prominences have radius $r \sim 5 \times 10^9$ cm (Serio *et al.* 1978) corresponding to $L \sim 8 \times 10^{10}$ cm, if we suppose that the coronal loops are in a configuration with the temperature maximum along the loop and the prominence material is the cool material near the temperature minimum, we find that for $s_H \lesssim 4 \times 10^9$ cm, loops will develop a temperature inversion which might correspond to "prominence" solutions.

The final stage of coronal loop evolution is attained when the defining coronal magnetic fields have sufficiently diffused that small local field perturbations (as might occur because of the emergence of new magnetic flux into the corona) suffice to "open" the field lines to the interplanetary medium. The removal of the constraint of plasma confinement drastically alters the structure of the now open atmosphere. In addition to allowing a new energy sink (the solar wind loss), the open topology also removes the previous connection of coronal pressure and temperature. This connection is lost because plasma is no longer confined and because the nature of mechanical heat deposition may change as a function of field topology.

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